

SQL CASE STUDY ON HR DATA

Q1---Total Employee

```
select count(*) as Total_employee  
from hr
```

total_employee
200

Q2---Avg Absent Days

```
select round(avg(absent_days),2) as avg_absent  
from hr
```

avg_absent
10.82

Q3---Active Employees

```
SELECT  
COUNT(*) AS active_count,  
'Active Employees' AS label  
FROM hr  
WHERE employment_status = 'Active';
```

active_count	label
130	Active Employee...

Q4---Avg Salary

```
select round(avg(annual_salary_eur),2) as avg_salary  
from hr
```

avg_salary
72797.00

Q5---City Wise Employee

```
select location, count(*) as number_of_emp  
from hr  
group by location;
```

location character varying (20)	number_of_emp bigint
Munich	48
Berlin	38
Stuttgart	34
Hamburg	46
Frankfurt	34

Q6--- Gender Wise Employee

- select gender, count(*) as emp
from hr
group by gender

gender character varying (10)	emp bigint
Female	101
Male	99

Q7--- Department Wise Salary

- select department,round(sum(annual_salary_eur),2) as total_salary
from hr
group by department
order by total_salary desc

department character varying (20)	total_salary numeric
Sales	2366700.00
Customer Support	2281700.00
Operations	2269400.00
Engineering	2148600.00
Finance	2030400.00
HR	1966100.00
Marketing	1496500.00

Q8---Attrition Rate

- SELECT
ROUND(
COUNT(CASE WHEN employment_status = 'Terminated' THEN 1 END) *
100.0) /
COUNT(*),
2
) AS attrition_rate
FROM hr;

attrition_rate numeric
18.00

Q9---Attrition Rate By Dept

```
- select department,ROUND(
                                (COUNT(CASE WHEN employment_status = 'Terminated'
                                THEN 1
                                END)*100.0)/
                                count(*),
                                2
                                ) as attrition_rate
from hr
group by department
```

department	attrition_rate
character varying (20)	numeric
Finance	28.57
Operations	26.67
HR	17.86
Sales	20.00
Engineering	6.67
Customer Support	18.75
Marketing	4.55

Q10---Attrition Rate By Job Label

```
select job_level,round(
                                (count(case when employment_status='Terminated' then 1
                                end)*100.0)/
                                count(*),
                                2
                                )as Attrition_Rate
from hr
group by job_level
```

job_level	attrition_rate
character varying (20)	numeric
Manager	17.31
Senior	20.93
Lead	20.93
Junior	14.71
Mid	14.29

Q11---Total Resigned By Performance

```
select performance_rating,
count(case when employment_status='Resigned' then 1 end) As total_resigned
from hr
group by performance_rating
```

performance_rating	total_resigned
character varying (15)	bigint
Below Average	11
Average	6
Good	8
Excellent	9

Q12---Total Resigned By Year

```

select To_char(exit_date,'YYYY')as year,
count(case when employment_status='Resigned' then 1 end) As total_resigned
from hr
where exit_date is not null
group by To_char(exit_date,'YYYY')
order by year

```

year	total_resigned
2020	2
2021	4
2022	8
2023	6
2024	8
2025	3
2026	2

Q13---Top Resigned Employee By Tenure

```

SELECT
    employee_id,
    name,
    department,
    employment_status,
    ROUND(EXTRACT(YEAR FROM AGE(exit_date, hire_date))
    + EXTRACT(MONTH FROM AGE(exit_date, hire_date)) / 12, 2) AS tenure_years
FROM hr
WHERE employment_status = 'Resigned' and exit_date is not null
ORDER BY tenure_years DESC
LIMIT 1;

```

employee_id	name	department	employment_status	tenure_years
[PK] text	character varying (20)	character varying (20)	character varying (10)	numeric
EMP0166	Employee_166	Sales	Resigned	4.00

Q14--Total Resigned

```

select count(case when employment_status='Resigned' then 1 end) as total_resigned
from hr

```

total_resigned
bigint
34

Q15---Average Tenure Exit

```

select employment_status,
round(avg(extract(year from age(exit_date, hire_date))
+extract(month from age(exit_date, hire_date))/12),2) as avg_tenure_year
from hr
where employment_status='Resigned'
group by employment_status

```

employment_status	avg_tenure_year
character varying (10)	numeric
Resigned	2.51

Q16---Avg Satisfaction Score Of Resigned Employee

```
select employment_status,
round(avg(job_satisfaction_score),2) as avg_satisfaction
from hr
where employment_status='Resigned'
group by employment_status
```

employment_status	avg_satisfaction
character varying (10)	numeric
Resigned	3.14

Q17---Average Salary

```
select round(avg(annual_salary_eur),2) as avg_salary
from hr
```

avg_salary
numeric
72797.00

Q18---Min Salary

```
select min(annual_salary_eur) as min_salary
from hr
```

min_salary
numeric
35300

Q19---Max Salary

```
select max(annual_salary_eur) as max_salary
from hr
```

max_salary
numeric
109900

Q20---Average Salary By Education Level

```
select education, round(avg(annual_salary_eur),2) as avg_salary
from hr
group by education
```

education character varying (15) 🔒	avg_salary numeric 🔒
Master	75156.10
Bachelor	71948.21
PhD	72367.44
Diploma	72285.00

Q21---Average Salary By Department

```
select department, round(avg(annual_Salary_eur),2) as avg_salary
from hr
group by department
```

department character varying (20) 🔒	avg_salary numeric 🔒
Finance	72514.29
Operations	75646.67
HR	70217.86
Sales	78890.00
Engineering	71620.00
Customer Support	71303.13
Marketing	68022.73

Q22---Average Salary By Job Level

```
select job_level, round(avg(annual_Salary_eur),2) as avg_salary
from hr
group by Job_level
```

job_level character varying (20) 🔒	avg_salary numeric 🔒
Manager	97182.69
Senior	69739.53
Lead	85113.95
Junior	40064.71
Mid	53035.71

Q23---Average Salary By Gender

```
select gender, round(avg(annual_Salary_eur),2) as avg_salary
from hr
group by gender
```

gender character varying (10) 🔒	avg_salary numeric 🔒
Female	73820.79
Male	71752.53

Q24---Average Salary By Year

```
select extract(year from hire_date) as year, round(avg(annual_Salary_eur),2) as
avg_salary
from hr
```

group by year
order by year

year	avg_salary
numeric	numeric
2018	72968.57
2019	72678.79
2020	64465.63
2021	80131.25
2022	69808.70
2023	80400.00

Q25---Salary By Performance Ratio

```

SELECT
  employee_id,
  ROUND(annual_salary_eur, 2) AS salary,
  performance_rating,
  ROUND(
    annual_salary_eur /
    CASE
      WHEN performance_rating = 'Excellent' THEN 5
      WHEN performance_rating = 'Very Good' THEN 4
      WHEN performance_rating = 'Good' THEN 3
      WHEN performance_rating = 'Average' THEN 2
      WHEN performance_rating = 'Poor' THEN 1
      ELSE NULL
    END,
    2) AS salary_performance_ratio
FROM hr
WHERE performance_rating IS NOT NULL
ORDER BY salary_performance_ratio DESC;

```

employee_id	salary	performance_rating	salary_performance_ratio
[PK] text	numeric	character varying (15)	numeric
EMP0100	96900.00	Average	48450.00
EMP0130	95500.00	Average	47750.00
EMP0134	95000.00	Average	47500.00
EMP0198	94900.00	Average	47450.00
EMP0113	94600.00	Average	47300.00
EMP0179	94400.00	Average	47200.00
EMP0079	94000.00	Average	47000.00
EMP0136	92700.00	Average	46350.00
EMP0127	91600.00	Average	45800.00
EMP0090	90400.00	Average	45200.00

Q26---Average Training Hour

```
select round(avg(training_hours),2) as avg_training_hours
from hr
```

avg_training_hours
numeric
43.19

Q27---Giving Promotion Which Employee Is Working More Then Seven Years

```
select employee_id,
Name,
hire_date,
annual_salary_eur,
extract(year from age(current_date, hire_date)) as years_with_company
from hr
where age(current_date,hire_date)>'7 year'
order by years_with_company desc
limit 5
```

employee_id [PK] text	name character varying (20)	hire_date date	annual_salary_eur numeric	years_with_company numeric
EMP0051	Employee_51	2018-01-29	63600	8
EMP0068	Employee_68	2018-05-21	85900	8
EMP0025	Employee_25	2018-03-27	51400	8
EMP0038	Employee_38	2018-05-08	96600	8
EMP0069	Employee_69	2018-04-25	38200	8

Q28---Average Absent Days

```
select round(avg(absent_days),2) as avg_absent
from hr
```

avg_absent
numeric
10.82

Q29---Average Training_Hours

```
select round(avg(training_hours),2) as Avg_training_hours
from hr
```

avg_training_hours
numeric
43.19

Q30---Average Training Hours By Department

```
select department,round(avg(training_hours),2) as Avg_training_hours
from hr
group by department
```


department character varying (20)	avg_training_hours numeric
Finance	51.93
Operations	36.20
HR	43.18
Sales	43.53
Engineering	43.43
Customer Support	45.84
Marketing	36.91

Q31--Avg Absent Days By Department

```
select department,round(avg(absent_days),2) as avg_absent
from hr
group by department
```

department character varying (20)	avg_absent numeric
Finance	10.82
Operations	12.17
HR	10.93
Sales	8.47
Engineering	11.10
Customer Support	11.63
Marketing	10.45

Q32---Average Training Hours By Performance

```
select performance_rating,round(avg(training_hours),2) as Avg_training_hours
from hr
group by performance_rating
```

performance_rating character varying (15)	avg_training_hours numeric
Below Average	43.51
Average	41.31
Good	44.98
Excellent	43.05

Q33---Department Wise Performance

```
select department, performance_rating ,count(*) as total_count
from hr
group by department, performance_rating
order by department, total_count desc
```

department	performance_rating	total_count
character varying (20)	character varying (15)	bigint
Customer Support	Good	10
Customer Support	Below Average	10
Customer Support	Average	8
Customer Support	Excellent	4
Engineering	Good	10
Engineering	Excellent	9
Engineering	Below Average	6
Engineering	Average	5
Finance	Average	10
Finance	Good	7
Marketing	Good	5
Marketing	Average	5
Operations	Below Average	9
Operations	Good	7
Operations	Average	7
Operations	Excellent	7
Sales	Below Average	11
Sales	Good	8
Sales	Excellent	7
Sales	Average	4

department	performance_rating	total_count
character varying (20)	character varying (15)	bigint
Finance	Below Average	6
Finance	Excellent	5
HR	Average	16
HR	Below Average	4
HR	Good	4
HR	Excellent	4
Marketing	Below Average	7
Marketing	Excellent	5
Marketing	Good	5
Marketing	Average	5

Q34---Total Hires

```
select count(*) as total_hires
from hr
```

total_hires
bigint
200

Q35---Top Job Sources

```
select recruitment_source, count(*) as total
from hr
group by recruitment_source
order by total desc
limit 3
```

recruitment_source	total
character varying (15)	bigint
Company Website	49
LinkedIn	43
Indeed	38

Q36---Most Hire By Department

```
select department, count(*) as total_hired
from hr
group by department
order by total_hired desc
```

department	total_hired
character varying (20)	bigint
Customer Support	32
Operations	30
Sales	30
Engineering	30
HR	28
Finance	28
Marketing	22

Q37---Most Hired By Job Level

```
select job_level, count(*)as total_hired
from hr
group by job_level
order by total_hired desc
```

job_level character varying (20)	total_hired bigint
Manager	52
Senior	43
Lead	43
Junior	34
Mid	28

Q38---Hired By Education Level

```
select education , count(*)as total_hired
from hr
group by education
order by total_hired desc
```

education character varying (15)	total_hired bigint
Diploma	60
Bachelor	56
PhD	43
Master	41

Q39---Hired By Year

```
select extract(year from hire_date) as year ,count(*) as total_hired
from hr
group by year
order by year
```

year numeric	total_hired bigint
2018	35
2019	33
2020	32
2021	32
2022	46
2023	22